



وزارة الاتصالات
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SUNNOVA

SOLAR POWER GENERATION FORECASTING

Machine Learning and Deep Learning for 15-Minute and 24-Hour-Ahead Photovoltaic Forecasting

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INTRODUCTION

Solar Power Forecasting: Context and Need

RENEWABLE ENERGY ENABLES A SUSTAINABLE POWER FUTURE

- Renewable energy comes from naturally replenished sources such as solar, wind, and hydropower.
- Solar power is one of the fastest-growing renewable sources because it is clean, scalable, and suitable for utility-scale generation.
- However, its variability creates a strong need for accurate power forecasting.

Forecasting makes renewable power
more predictable and easier to integrate into the grid.



SOLAR PV CONVERTS SUNLIGHT INTO ELECTRICITY

- Solar photovoltaic systems convert sunlight directly into electrical power.
- Their output depends mainly on solar irradiance, temperature, system characteristics, and local weather conditions.
- PV systems are used in residential, commercial, industrial, and utility-scale applications.



Solar energy
is abundant, but its power output is highly variable and weather-dependent.

FORECASTING MAKES SOLAR POWER GRID-READY

- Solar power forecasting estimates future photovoltaic generation using historical power, weather, and solar-position information.
- Accurate forecasts help grid operators balance supply and demand, schedule energy resources, reduce curtailment, and integrate renewable generation more reliably.

**Better forecasts
lead to better energy planning and more reliable renewable integration.**



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MOTIVATION

Why Solar Forecasting Matters

WEATHER VARIABILITY MAKES SOLAR FORECASTING DIFFICULT

Cloud Cover

Causes sudden irradiance drops and rapid power fluctuations.

Temperature

High temperature can reduce PV conversion efficiency.

Humidity & Solar Irradiance

Affects atmospheric light transmission and irradiance quality.

Grid Stability & Efficiency

Forecast accuracy supports balancing, scheduling, and reliable renewable integration.

“ The forecasting model must combine weather dynamics with historical power behavior ”

WHY MACHINE LEARNING FOR SOLAR FORECASTING?

Learning nonlinear relationships between weather conditions and photovoltaic power generation

Nonlinear Modeling

Captures complex relationships between weather, solar position, and PV output.

Multi-Source Features

Combines weather data, historical power, solar geometry, and plant metadata.

Multi-Horizon Forecasting

Supports 15-minute-ahead and 24-hour-ahead solar power prediction.

Operational Value

Supports energy scheduling, grid integration, and performance monitoring.

Machine learning transforms heterogeneous solar data into actionable power forecasts.

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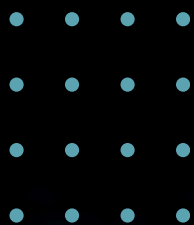
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CHALLENGES

Technical Obstacles in Solar Power Forecasting



PROBLEM STATEMENT: SOLAR OUTPUT IS HARD TO PREDICT



- Solar generation changes across time and seasons.
- Sudden weather changes create rapid power fluctuations.
- Historical power alone is insufficient under changing weather.
- Forecast accuracy decreases without weather-feature integration.

“Reliable forecasting requires both temporal history and weather information.”

WEATHER VARIABILITY IS A MAJOR SOURCE OF FORECAST UNCERTAINTY

Cloud Cover

- Blocks or scatters solar radiation and can cause rapid drops in PV output.

Temperature & Humidity

- Affect panel efficiency and the transmission of sunlight through the atmosphere.

Solar Irradiance

- Varies with time, season, location, and atmospheric conditions; it is the primary driver of PV generation.

“ The model must integrate weather features with historical power to capture these variations. ”

SOLAR GENERATION HAS COMPLEX TEMPORAL PATTERNS

Power output depends on both recent history and long-term cycles.

Daily Cycles

Generation rises during daylight and falls to zero at night.

Seasonal Patterns

Irradiance and production vary across seasons.

Rapid Changes

Cloud shadows can create sudden intra-hour power drops.

Long-Term Effects

Panel degradation and changing operating conditions affect generation over time.

The model must capture patterns across multiple time scales.

FORECAST ERRORS CREATE OPERATIONAL AND ECONOMIC RISK



- **Grid Imbalance** — Inaccurate forecasts complicate supply–demand balancing.
- **Poor Scheduling** — Operators cannot plan dispatch and storage efficiently.
- **Higher Costs** — Forecast errors may increase reserve requirements and energy waste.

“ The operational value of a forecasting model is measured by how well it supports real grid decisions. ”

AI MODELS LEARN COMPLEX WEATHER-POWER PATTERNS

Nonlinear Relationships

**Captures interactions between
weather and PV output**

Feature Integration

**Combines weather, historical
power, and solar-position
features**

Operational Support

**Produces forecasts for scheduling
and grid balancing**

These capabilities motivate the use of Random Forest and LSTM in the proposed solution.

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RELATED WORK

Previous Research in Solar Power Forecasting

SELECTED STUDIES IN SOLAR POWER FORECASTING

Study	Dataset / Horizon	Model	Reported Result	Main Limitation
Konstantinou et al. (2021)	Cyprus PV plant; 1.5-hour ahead	Stacked LSTM	RMSE = 0.11368	Single plant and limited horizon
Jesus (2025)	BR-PVGen; 51 plants	MLP + TL / RF	SMAPE = 43.92% NRMSE = 14.07% / 13.82%	Static models limited temporal learning
Radzi et al. (2026)	Malaysian PV plant; 38 months	Hybrid GRU–LSTM	$R^2 = 0.9961$ RMSE = 18.55 W MAE = 19.43 W	Higher complexity and deployment cost

Results are not directly comparable because datasets, forecast horizons, units, and evaluation protocols differ.

These differences motivate a consistent multi-plant, multi-horizon evaluation.

RESEARCH GAP

A consistent multi-plant, multi-horizon evaluation is still needed.

Limited Generalization

Many studies focus on a single PV plant or a limited operating condition.

Horizon Dependency

A model that performs well at 15 minutes may not perform equally well at 24 hours.

Model Trade-Offs

Random Forest is interpretable and robust, while LSTM models temporal dependencies but requires longer sequences.

Fair Evaluation

A consistent dataset, plant-level split, feature pipeline, and evaluation metrics are needed for meaningful comparison.

This project evaluates Random Forest and LSTM across 51 PV plants and two forecasting horizons.

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PROJECT GOAL

RELIABLE MULTI-HORIZON FORECASTING

- Enable smarter energy trading decisions by providing accurate day-ahead solar power forecasts that allow plant operators and grid managers to reduce reserve costs and maximise revenue from electricity market participation
- Design a generalised forecasting framework that adapts to new unseen plants using plant metadata (capacity, efficiency, tracker type, location) without requiring retraining from scratch
- Implement Transfer Learning to improve prediction accuracy for plants with limited historical data by leveraging knowledge from data-rich plants
- Deploy the trained models through a web application that allows users to input plant specifications and weather forecasts and receive predicted power output curves for any plant — known or new

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PROPOSED SOLUTION

END-TO-END FORECASTING PIPELINE

1. **Data Preparation** — Clean, synchronize, and resample power and weather data.
2. **Feature Engineering** — Generate solar geometry, lag, rolling, temporal, and metadata features.
3. **Model Training** — Train per-plant Random Forest and LSTM models.
4. **Forecast Generation** — Produce 15-minute-ahead and 24-hour-ahead predictions.
5. **Evaluation & Comparison** — Compare models using R^2 , MAE, RMSE, NRMSE, and SMAPE.

RANDOM FOREST: AN INTERPRETABLE BASELINE

Random Forest combines multiple decision trees and averages their predictions to produce a stable regression output.

HOW RANDOM FOREST PRODUCES A FORECAST:

- 1. Sample — Train each tree on a different randomized subset of the data.**
- 2. Predict — Each tree produces an individual power forecast.**
- 3. Aggregate — Average the tree predictions to obtain the final output.**

KEY STRENGTHS:

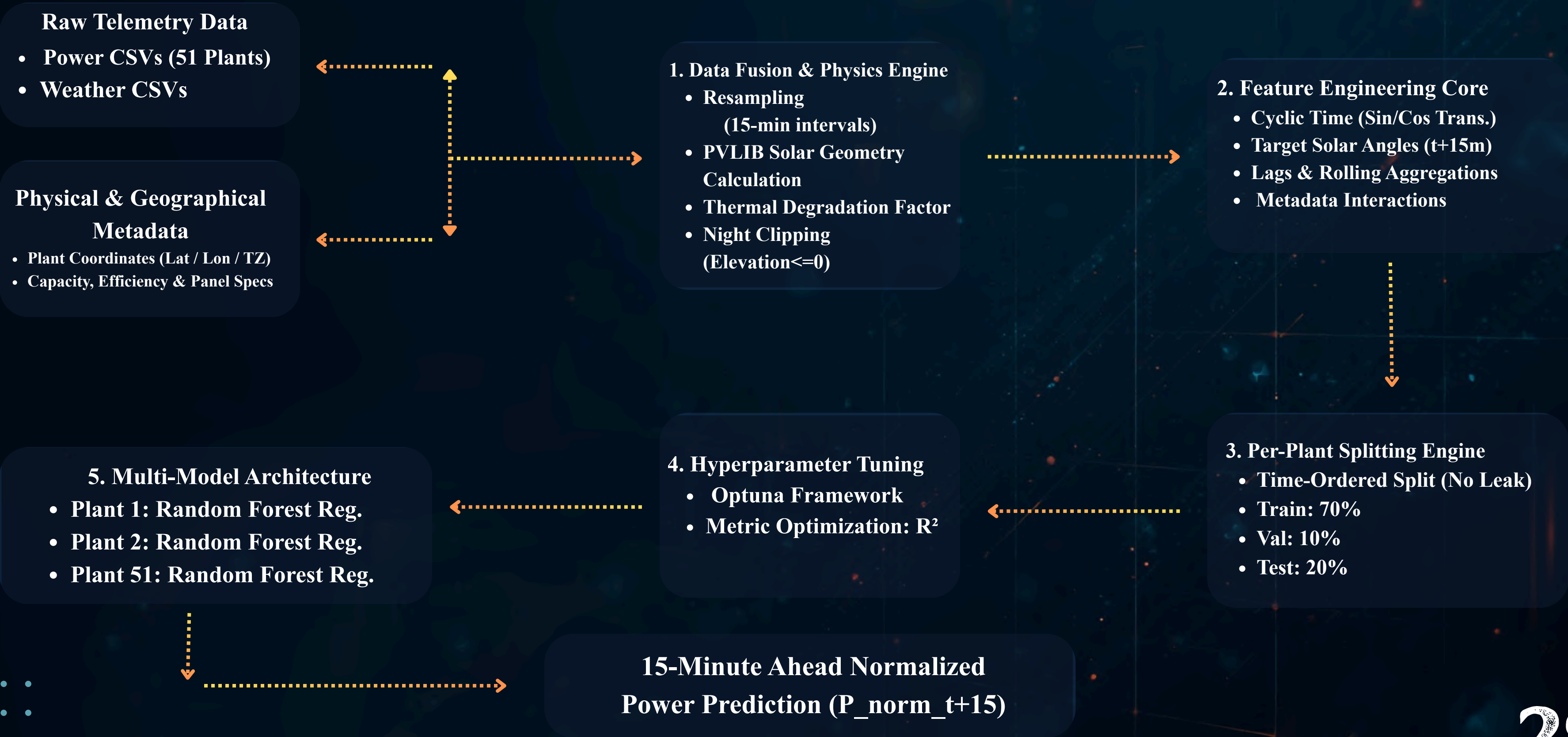
It captures nonlinear relationships and provides feature importance for interpretation.

WHY RANDOM FOREST FOR SOLAR FORECASTING?

- **Nonlinear Relationships** — Models complex interactions between weather and PV output.
- **Mixed Feature Types** — Uses weather, historical power, solar geometry, and plant metadata.
- **Interpretability** — Provides feature importance to explain forecast drivers.
- **Strong Baseline** — Offers a stable reference before evaluating sequential LSTM models.

SOLAR POWER FORECASTING SYSTEM (15-MINUTE AHEAD)

RF- Architecture:



SOLAR POWER FORECASTING SYSTEM (24-HOURE AHEAD)

RF- Architecture:

Raw Telemetry Data

- Power CSVs (51 Plants)
- Weather CSVs

Physical & Geographical Metadata

- Plant Coordinates (Lat / Lon / TZ)
- Capacity, Efficiency & Panel Specs

1. Paring & integrity

- ID- based power - weather pairing
- 51 matched plant profiles
- Meta data and folder validation

2. Physics - Informed Fusion

- Timezone normaliztion and UTC alignment
- Power/Weather inner merge
- 15 min resampling and interpolation
- Pvlib solar elevation / azimuth
- Night Clipping: Elevation ≤ 0

3. Feature Engineering Core

- 49 features per plant
- Cyclical time + $t+24h$ solar geometry
- Power/Weather lags: 1h-7d
- Rolling statistics and interactions
- Target = Power shifted - 96 steps.

5. Evaluation & Diagnostics

- R^2 , MAE, RMSE , NRMSE , SMAPE
- Pooled Vs per- plant metrics.
- Outlier detection
- Feature importance and error profiles
- Maintenance dispatch report.

5. Inference & Post-Processing

- Per-plant prediction +surrogate fallback
- Inverse scaling to watts
- Non-negative clipping
- Target-Time night mask

4- Chronological split & RF

- Per - plant 80% train / 20% test
- RFRegressor per plat
- full/ light parameter tier.
- 51 dedicated model piplines

24 - Houre Ahead Power Forecast
Per Plant + Fleet

FEATURE ENGINEERING

Feature groups used to represent solar, weather, and temporal behavior

Physics-Informed Features

- Solar position (azimuth, zenith)
- Clear-sky index (pvlib)
- Solar geometry interactions

Lag Features

- Previous PV generation values
- Lagged meteorological variables
- Short-term temporal patterns

Plant Metadata

- Structure type (Tracker / Fixed)
- Bifaciality coefficient
- Installed capacity

Temporal & Weather

- Hour, day, and seasonal indicators
- Irradiance (POA / GHI)
- Temperature & related variables

RANDOM FOREST PIPELINE

CORE APPROACH

Model: Per-Plant Random Forest

Horizon: 15Min | 24H ahead

Dataset: BR-PVGen (51 plants)

Scope: Individual model per plant +
shared feature schema

Key Design Choices

- Physics-informed features (pvlib solar geometry)
- Clear-sky index + interaction terms
- Plant metadata (tracker / bifacial / capacity)
- Night-time hard masking (physics constraint)
- Hermetic packaging for production deployment

PIPELINE STEPS

- 1 Data Sync**
Power + Weather pairing per plant
- 2 Feature Eng.**
Solar position, clear-sky, lags, metadata
- 3 Train RF**
Independent Random Forest per plant
- 4 Night Mask**
Force zero output outside solar hours
- 5 Package**
Serialize model + schema + scales

TRAINING STRATEGY

Per-Plant Modeling

- **One model per plant — Captures site-specific generation behavior.**
- **Shared feature schema — Keeps inputs consistent across all plants.**
- **Chronological evaluation — Preserves temporal order and avoids data leakage.**
- **Operational benefit — Supports plant-level forecasting and diagnostics.**

RANDOM FOREST

KEY HYPERPARAMETERS

Configuration used for the 15-minute-ahead RF experiment

422

n_estimators

Number of trees in the forest.
Higher value improves stability and
reduces variance.

18

max_depth

Maximum depth of each tree.
Limits complexity to control overfitting.

4

min_samples_split

Minimum samples required to split
an internal node. Prevents overly
specific splits.

11

min_samples_leaf

Minimum samples required at a leaf
node. Improves generalization
on noisy data.

0.3

max_features

Number of features considered at each
split. Reduces correlation
between trees.

RANDOM FOREST KEY HYPERPARAMETERS

Configuration used for the 24-hour-ahead RF experiment

150

n_estimators

Number of trees in the forest.
Higher value improves stability and
reduces variance.

10

max_depth

Maximum depth of each tree.
Limits complexity to control overfitting.

30

min_samples_split

Minimum samples required to split
an internal node. Prevents overly
specific splits.

12

min_samples_leaf

Minimum samples required at a leaf
node. Improves generalization
on noisy data.

sqrt

max_features

Number of features considered at each
split. Reduces correlation
between trees.

LSTM: MODELING TEMPORAL DEPENDENCIES

Definition:

LSTM is a recurrent neural network designed to learn patterns across ordered time-series observations.

Key Characteristics:

- Learns long-term temporal dependencies.
- Uses historical weather and power sequences.
- Models nonlinear temporal relationships.

Unlike a static model, LSTM uses the order and history of observations.

LONG SHORT-TERM MEMORY (LSTM)

15MIN/24 MIN AHEAD

Architecture:

Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 96, 18)	0	-
meta_input (InputLayer)	(None, 7)	0	-
lstm_1 (LSTM)	(None, 96, 128)	75,264	seq_input[0][0]
meta_dense_1 (Dense)	(None, 32)	256	meta_input[0][0]
layer_normalization (LayerNormalization)	(None, 96, 128)	256	lstm_1[0][0]
batch_normalization (BatchNormalization)	(None, 32)	128	meta_dense_1[0][0]
lstm_2 (LSTM)	(None, 64)	49,408	layer_normalization[0][0]
dropout (Dropout)	(None, 32)	0	batch_normalization[0][0]
layer_normalization_1 (LayerNormalization)	(None, 64)	128	lstm_2[0][0]
meta_dense_2 (Dense)	(None, 16)	528	dropout[0][0]
merge (Concatenate)	(None, 80)	0	layer_normalization_1[0][...] meta_dense_2[0][0]
fc_1 (Dense)	(None, 128)	10,368	merge[0][0]
dropout_1 (Dropout)	(None, 128)	0	fc_1[0][0]
fc_2 (Dense)	(None, 64)	8,256	dropout_1[0][0]
dropout_2 (Dropout)	(None, 64)	0	fc_2[0][0]
output (Dense)	(None, 1)	65	dropout_2[0][0]
relu_clamp (ReLU)	(None, 1)	0	output[0][0]

LONG SHORT-TERM MEMORY (LSTM)

Why LSTM ?

Challenges with Random Forest:

- Processes each sample independently
- Cannot capture temporal dependencies
- Performance decreases under rapidly changing weather conditions

Advantages of LSTM

- Learns sequential weather patterns
- Retains long-term historical information
- Better models nonlinear temporal relationships

Evaluate whether explicit temporal learning improves forecasting across the two horizons.

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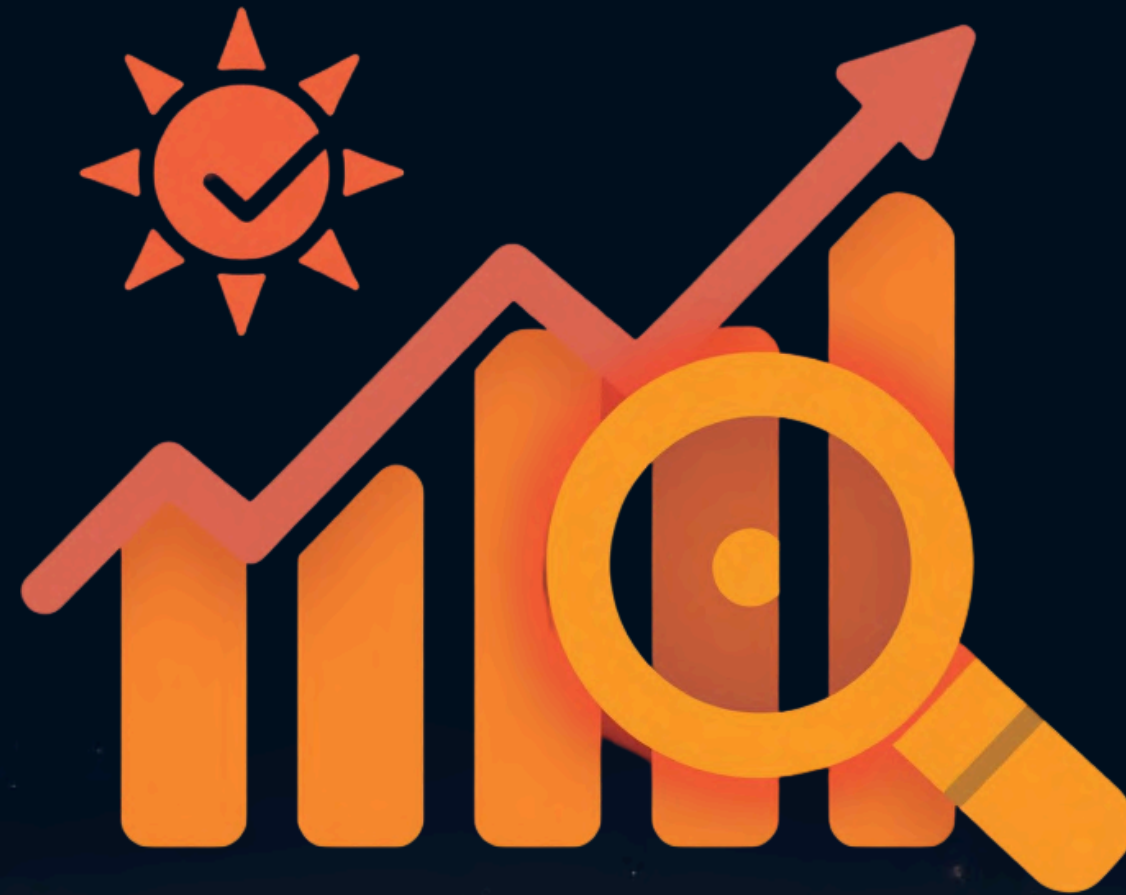
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EXPERIMENTAL RESULTS

Comparing Random Forest and LSTM
Across Two Forecast Horizons

RANDOM FOREST — 15-MINUTE-AHEAD RESULTS

0.8883

Mean R^2

50 plants evaluated

17.81

SMAPE (%)

Mean SMAPE
across 50 plants

10.52%

NRMSE (%)

Mean NRMSE
across all plants

KEY INSIGHTS

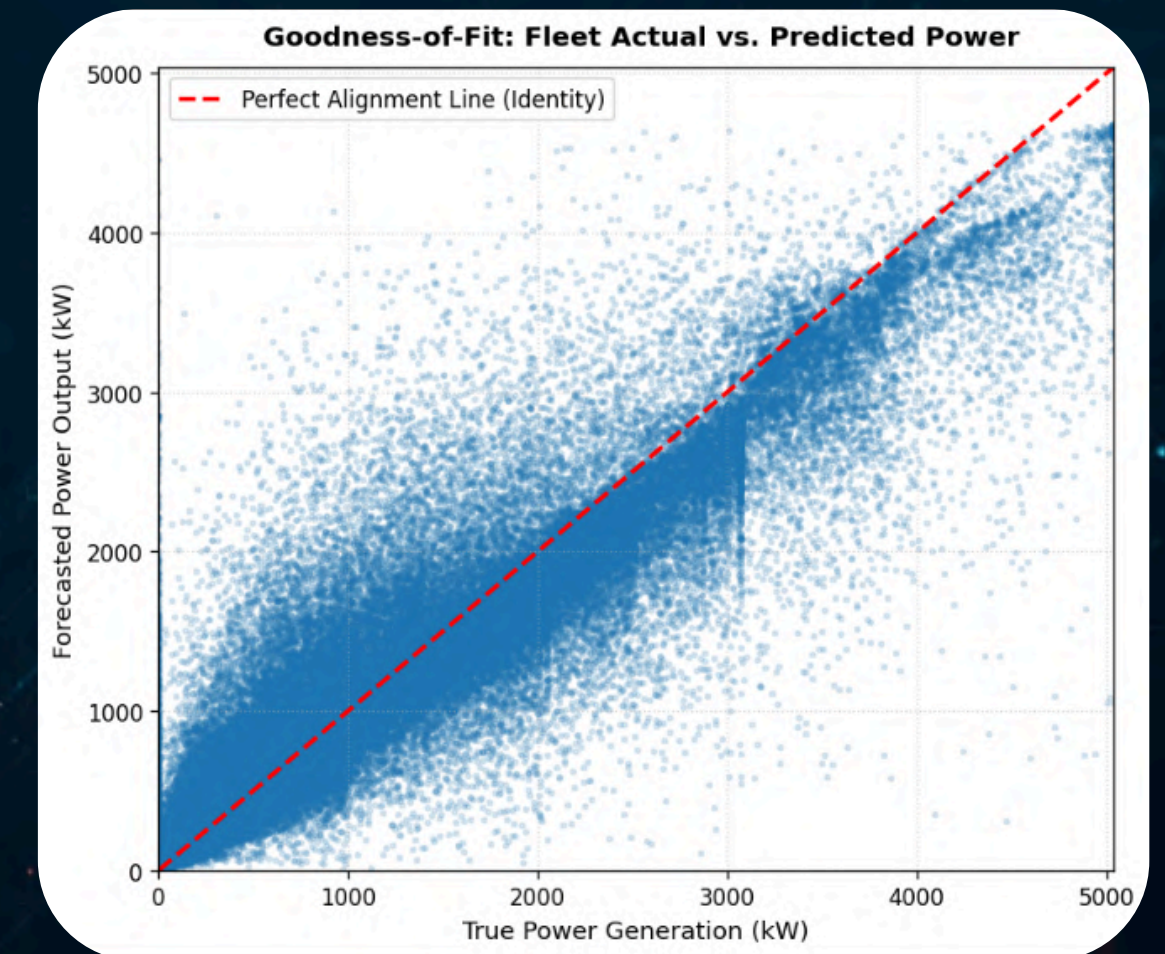
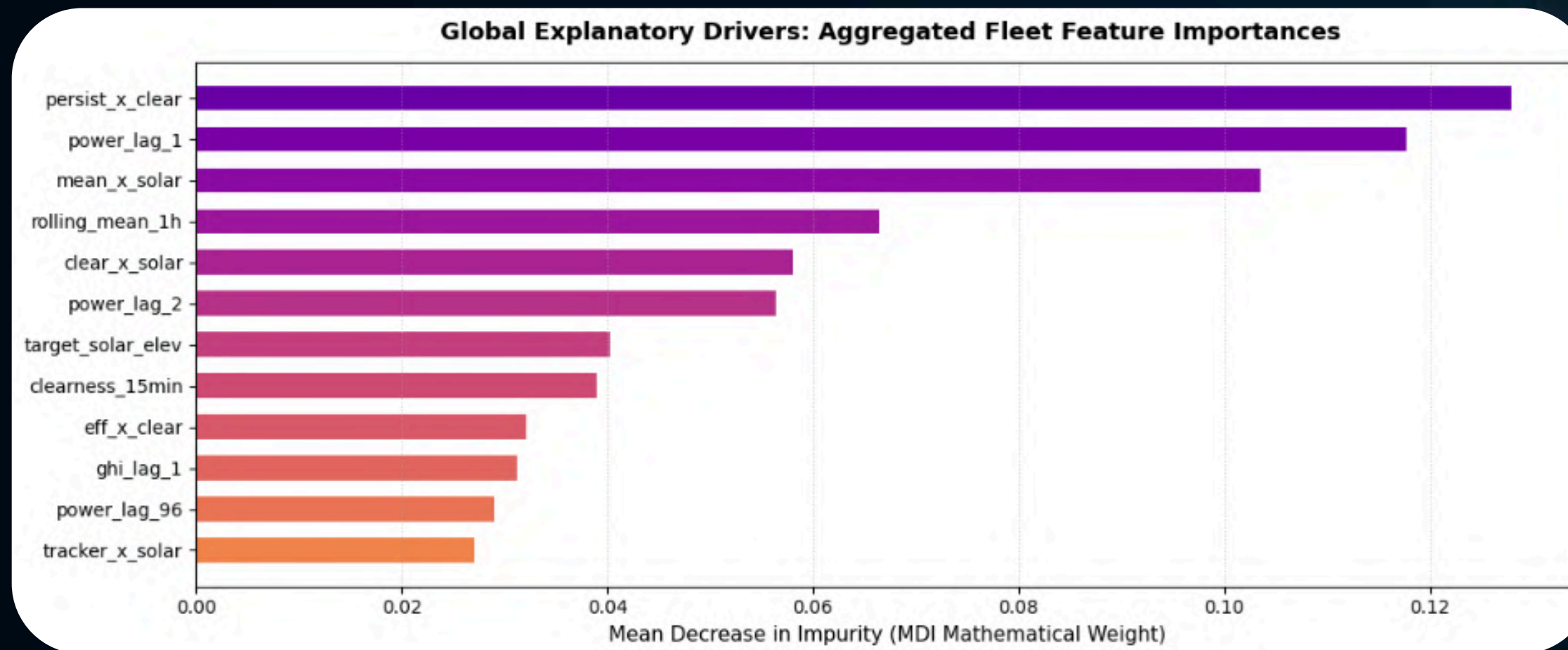
- 50 Plants evaluated (Plant 15 excluded)
- Best Validation $R^2 = 0.896$
- Most plants achieved $R^2 > 0.80$
- Errors increased during late afternoon

LIMITATIONS & NEXT STEP

- No temporal memory
- Peak power saturation

Next:
evaluate LSTM for temporal awareness.

RANDOM FOREST PREDICTION ANALYSIS



- Predictions closely follow actual power.
- Weather + historical power are the most influential features.
- Performance decreases during rapid weather changes.

RANDOM FOREST EXPERIMENT (2) - 24.H AHEAD RESULTS

0.782

Mean R^2

50 plants evaluated

22.5 %

SMAPE (%)

Mean SMAPE
across 50 plants

15.29 %

NRMSE (%)

Mean NRMSE
across 50 plants

KEY INSIGHTS

- 50 Plants evaluated after excluding plant 15
- SMAPE was lower than all six qualitative reference models.
- Pooled $R^2 = 0.81$; Per-plant $R^2 = 0.782$
- Several plants exceeded $R^2 > 0.85$

LIMITATIONS & NEXT STEP

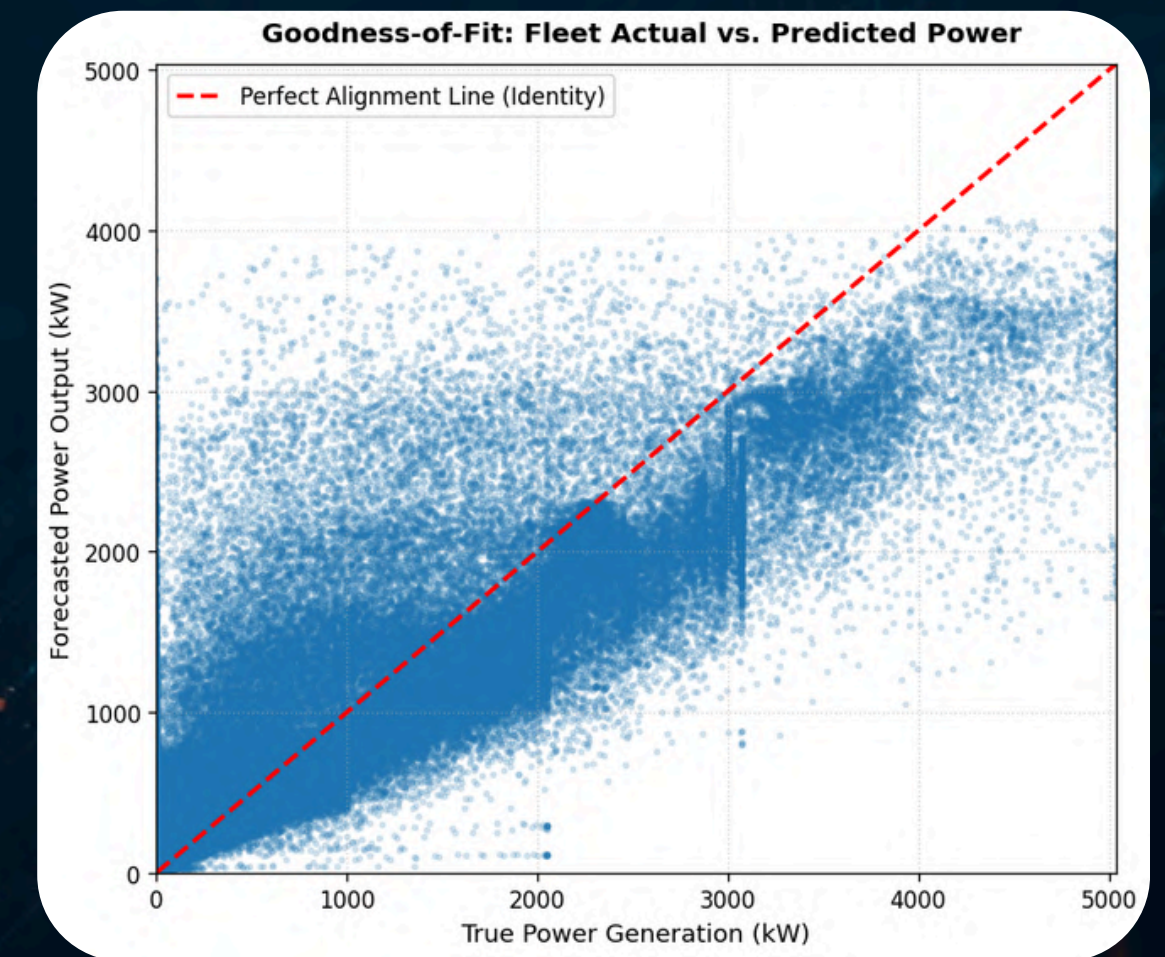
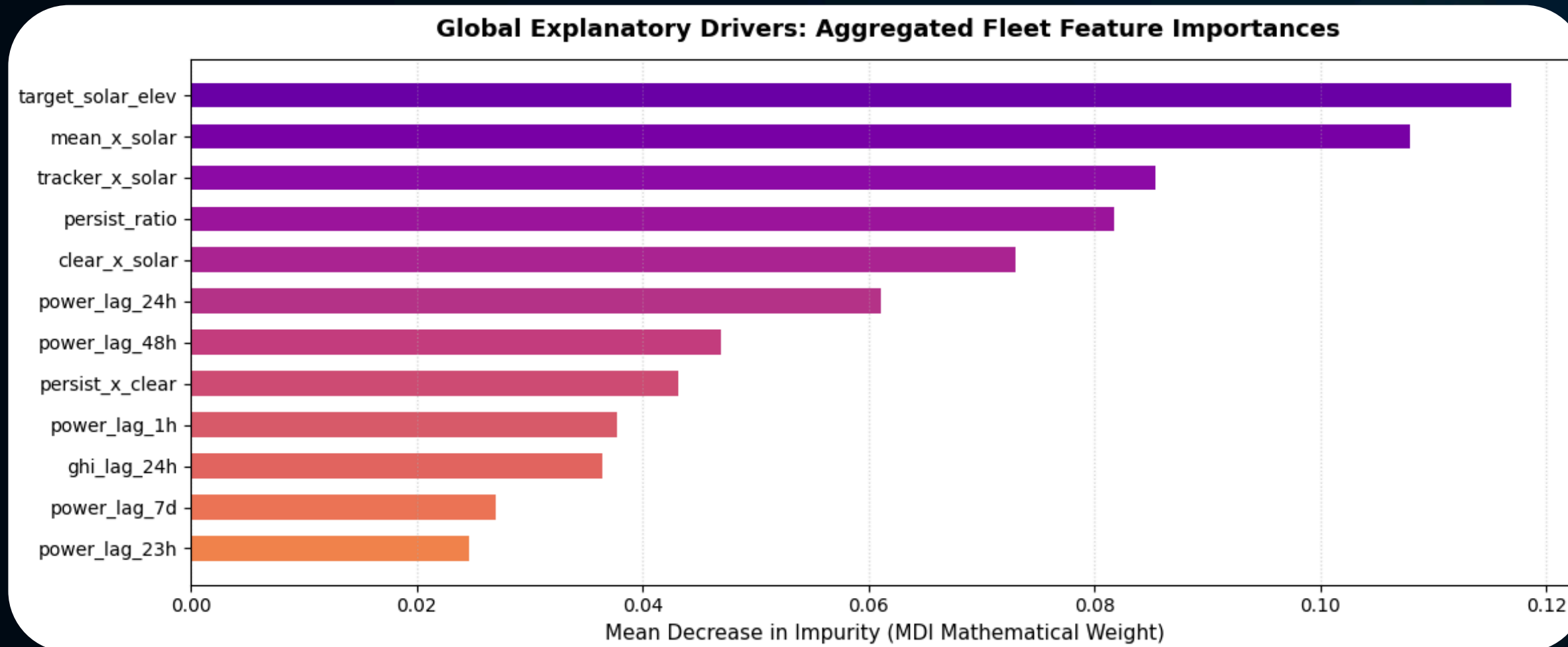
- 24-Hour Forecasting is harder than the 15-min experiment.
- Plant level Performance is heterogeneous ; plant 15 was a sever outlier
- Current RF has no explicit temporal dynamics

Next:

Compare LSTM / GRU for Temporal dynamics

RANDOM FOREST PREDICTION ANALYSIS EXPERIMENT 2 - 24.H

Explainability and Goodness-Of-Fit Across the Healthy Fleet



- Solar elevation is the dominant fleet-wide explanatory feature.
- 24h/48h persistence and engineered interactions reinforce the forecast.
- Extreme peaks are capped above approximately 3,500 kW, while cloud attenuation increases variance.

BASELINE COMPARISON: RESEARCH PAPER VS OUR RANDOM FOREST

Model	SMAPE (%)	NRMSE (%)
ARIMA	47.05	14.23
Gradient Boosting	46.94	13.99
Random Forest	47.33	13.82
XGBoost	46.48	13.92
MLP	46.40	14.49
MLP + Transfer Learning	43.92	14.07
Our Random Forest 15 Min / 24H	17.81/22.5	10.52/15.29

Key Findings

Our RF achieves lower SMAPE and NRMSE than the listed 15-minute reference models, while the 24-hour result demonstrates performance under a more difficult horizon.

LSTM — 15-MINUTE-AHEAD RESULTS

0.905

Mean R^2

50 plants (excl. p15)

36.88%

SMAPE (%)

Mean SMAPE
across all plants

15.82%

NRMSE (%)

Mean NRMSE
across all plants

KEY INSIGHTS

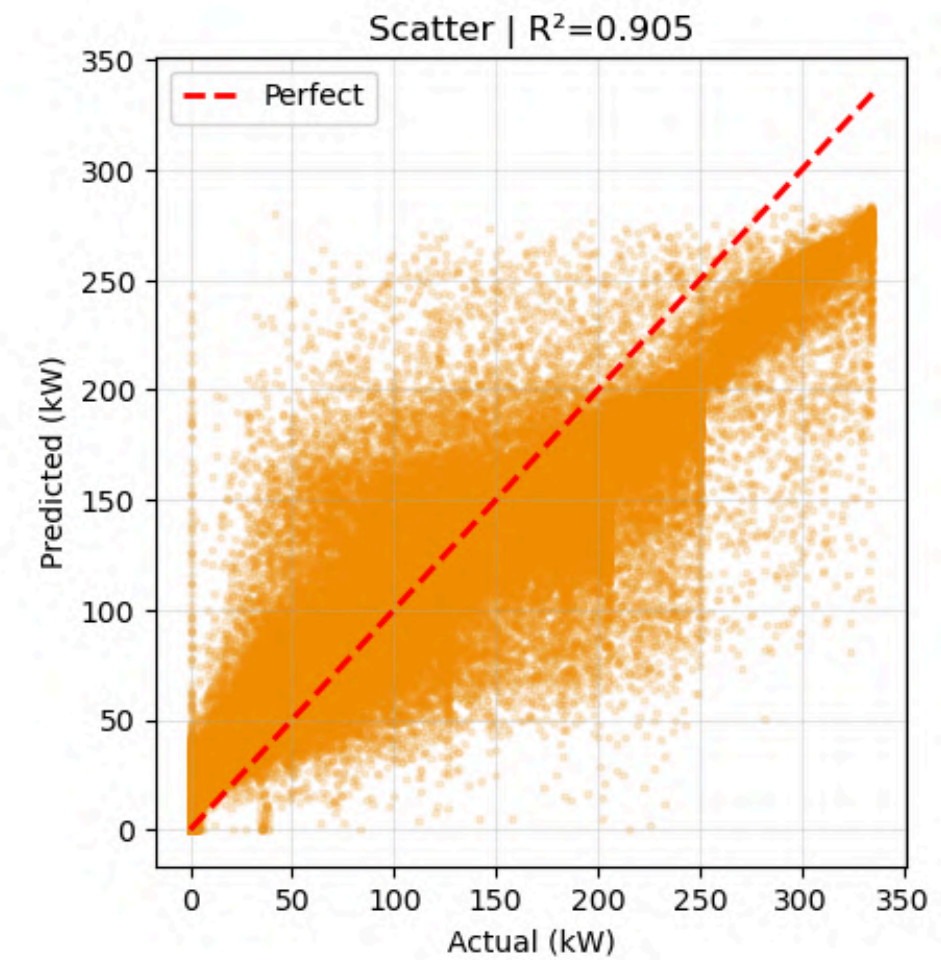
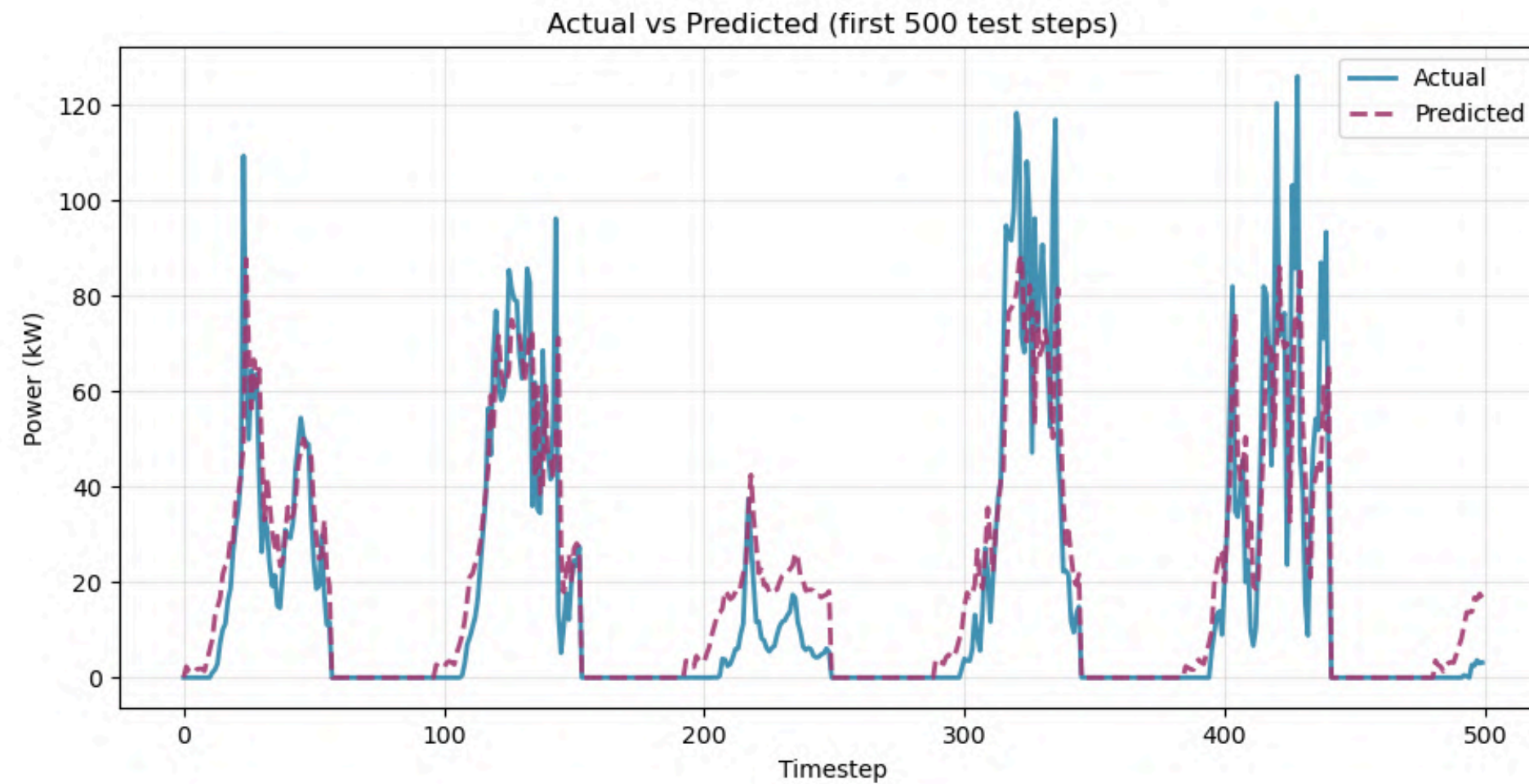
- 50 Plants evaluated (Plant 15 excluded)
- Mean Validation $R^2 = 0.905$
- Most plants achieved $R^2 > 0.80$
- LSTM explicitly models temporal dependencies.

LIMITATIONS

- Performance may decrease under highly variable weather.
- Results depend on sequence length.

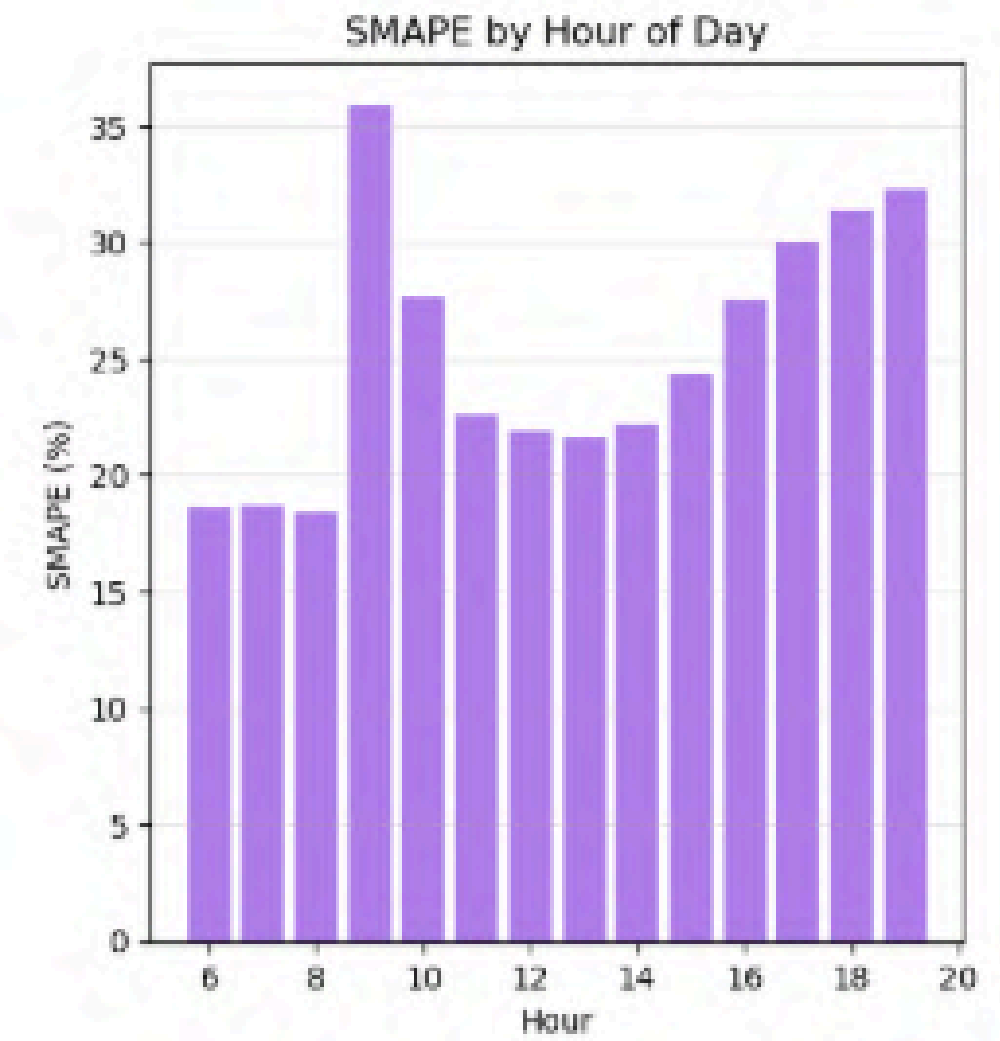
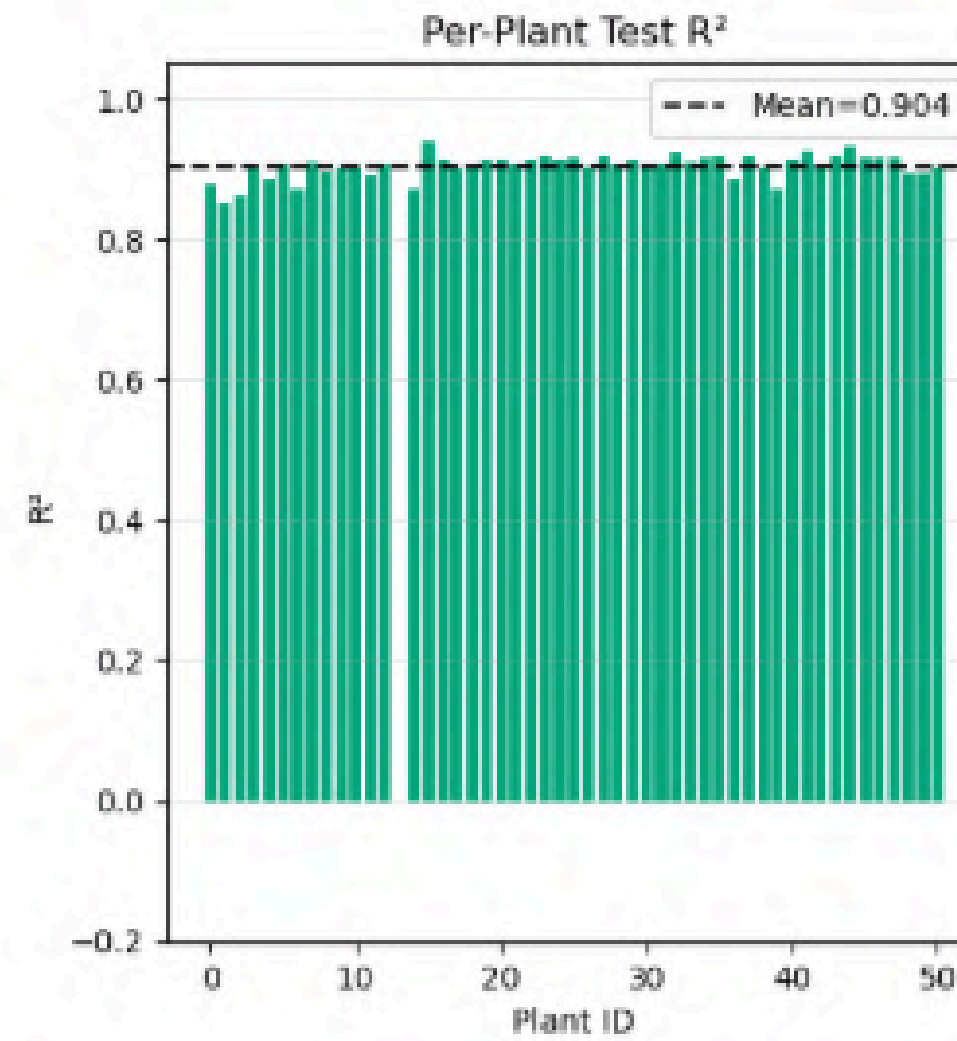
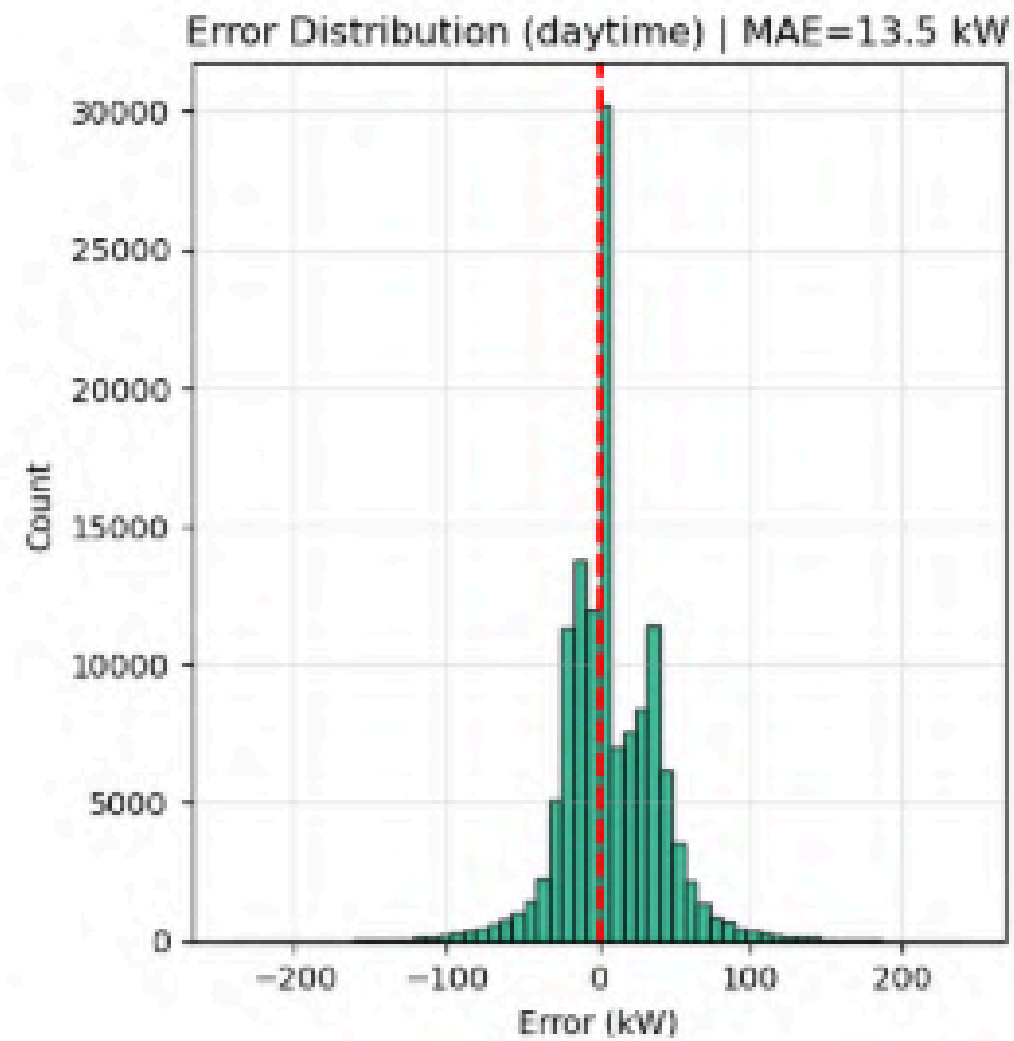
LSTM — 15-MINUTE-AHEAD PREDICTION ANALYSIS

LSTM Solar Forecast | 51 Plants | $R^2=0.905$ | SMAPE=23.4% | NRMSE(range)=7.7% | EXCELLENT



- **Trend Capture** — Predicted power follows the main daily generation pattern.
- **Forecast Agreement** — Predicted values show close agreement with actual power across the test sequence.
- **Weather Sensitivity** — Deviations increase during rapid weather changes.

LSTM 15 MIN- PREDICTION ANALYSIS



- Daytime errors are centered near zero.
- Performance is consistent across the evaluated plants.
- SMAPE increases during afternoon irradiance variability.

LSTM — 24-HOUR-AHEAD RESULTS

0.7919

Mean R^2

50 plants evaluated

33.30%

SMAPE (%)

Mean SMAPE
across 50 plants

29.64%

NRMSE (%)

Mean NRMSE
across 50 plants

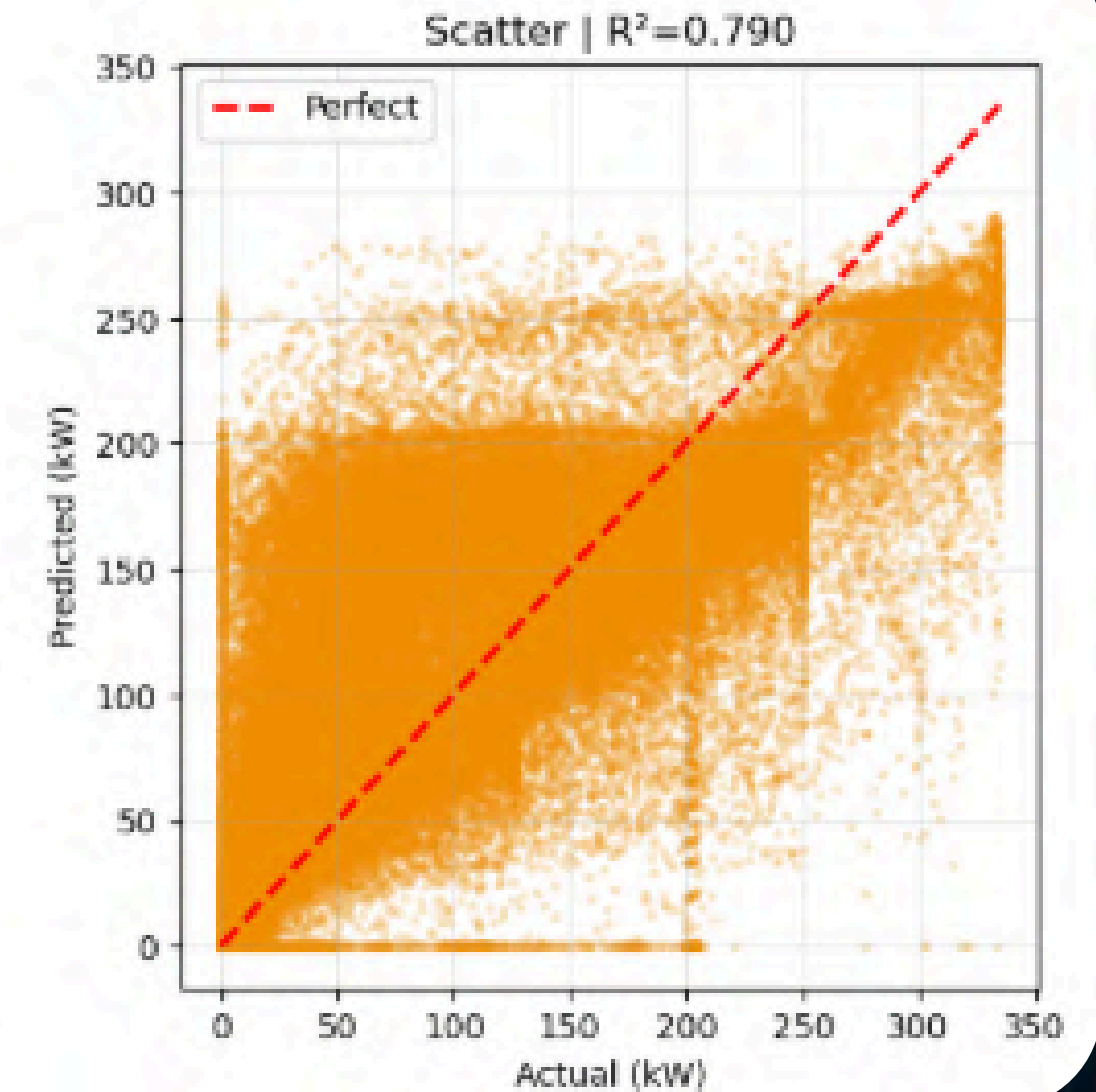
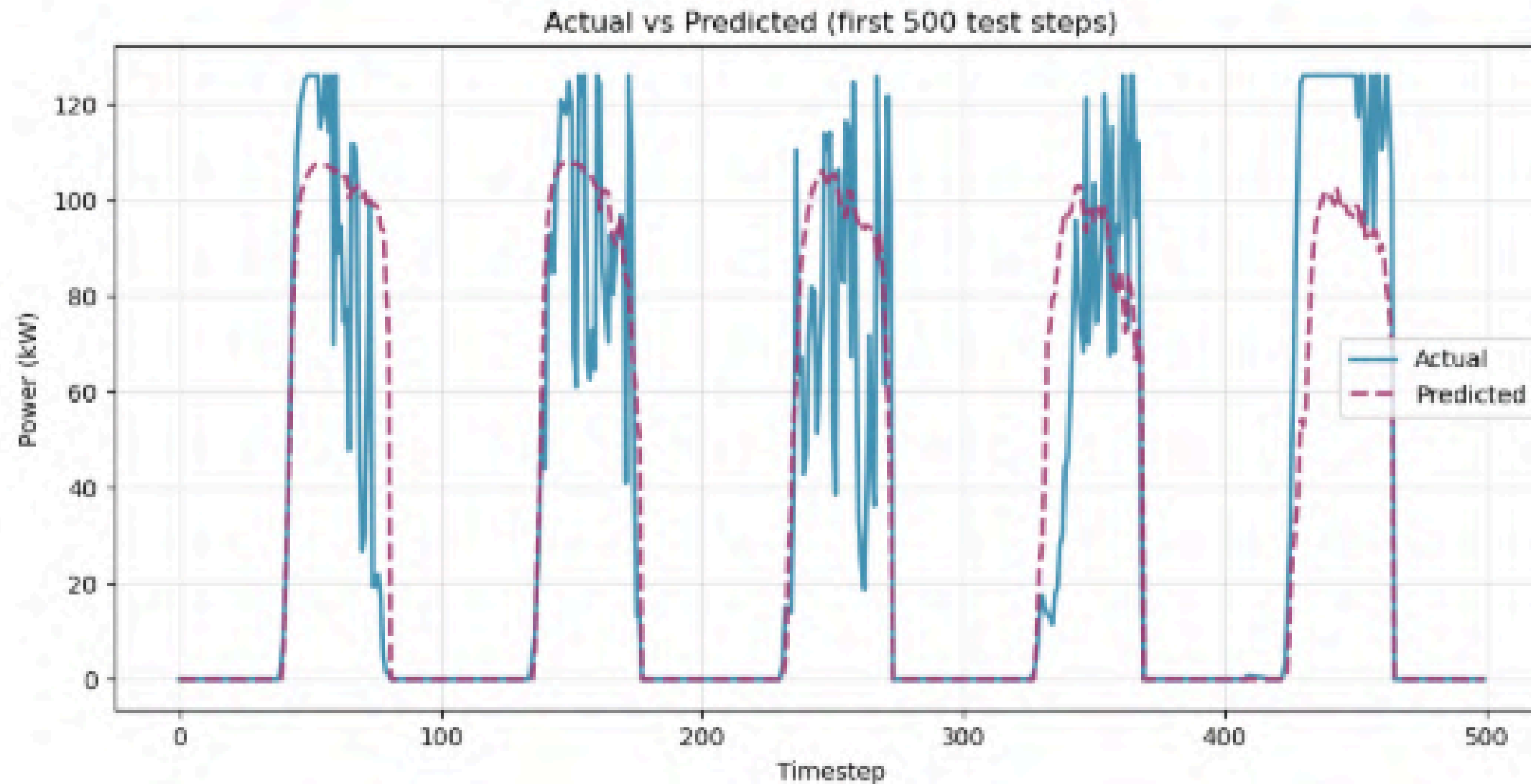
KEY INSIGHTS

- 50 Plants evaluated (Plant 15 excluded)
- Mean Test $R^2 = 0.7919$
- Improved overall accuracy compared with
Random Forest

LIMITATIONS

- Higher SMAPE at sunrise and sunset.
- The model cannot react immediately to
sudden weather shifts.
- Prediction requires 192 historical timesteps.

LSTM — 24-HOUR-AHEAD PREDICTION ANALYSIS



- **Pattern:** Captures the overall solar-generation pattern.
- **Peak Behavior:** Smooths sharp peaks and underestimates some rapid changes.
- **Performance:** Achieves a mean R^2 of approximately 0.79.

“LSTM captures the broad daily pattern,
but sharp peaks remain difficult to forecast 24 hours ahead.”

BASELINE COMPARISON: RESEARCH PAPER VS PROPOSED LSTM

Model	SMAPE (%)	NRMSE (%)
ARIMA	47.05	14.23
Gradient Boosting	46.94	13.99
Random Forest	47.33	13.82
XGBoost	46.48	13.92
MLP	46.40	14.49
MLP + Transfer Learning	43.92	14.07
Our LSTM 15 Min/24 H	36.88/33.30	15.82/29.64

Key Findings

- Mean per-plant $R^2 = 0.904$
- Better SMAPE than all reference models
- Same evaluation protocol as the reference paper
- 50 PV plants evaluated (Plant 15 excluded)

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CONCLUSION

- Developed an end-to-end solar power forecasting system using weather and historical power data.
- Built and compared per-plant Random Forest and LSTM models for 15-minute and 24-hour-ahead forecasting.
- Random Forest provided an interpretable and strong baseline, while LSTM added explicit temporal modeling.
- Model performance depends on the forecast horizon and evaluation metric.

“ The results show that model selection should match the operational horizon and the target performance metric. ”



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FUTURE WORK

- Hybrid RF + LSTM Models — Combine RF interpretability with LSTM temporal learning.
- Real-Time Forecasting — Deploy the pipeline for continuous operational predictions.
- Richer Data — Add weather features and evaluate larger datasets.
- Scalable Deployment — Extend Sunova toward cloud, smart-grid, and IoT integration.

“The next step is to turn the forecasting pipeline into a real-time operational tool.”



An aerial photograph of a vast solar farm at sunset. The rows of solar panels stretch towards the horizon, where a bright sun is setting, creating a warm orange glow. The sky is filled with soft, wispy clouds. The overall mood is peaceful and hopeful.

Thank You!

Questions & Discussion

